

The Reasonable Effectiveness of Using Linear Regression to Predict Student Performance in QC Introductory Mathematics Courses

Bryan Nevarez

Sunday 31st March, 2024

I. Introduction

A mathematics education at the college level is often accomplished by an institution implementing a wide variety of teaching styles and methods through their educators, namely professors, lecturers, adjuncts, with the goal of upholding a broad spectrum of academic standards with specified teaching outcomes for the university student. One can say that an overarching goal of an educator, particularly at the collegiate level, is to aptly convey collegiate level material to the student through a myriad of modalities such as lectures, labs, homework assignments, in-class assessments, etc., with the expectation that the student is, thus, sufficiently equipped with the necessary knowledge, skills, and experience to successfully continue their academic and/or professional careers.

On the other end, collegiate students have a multifarious set of data associated with them such as, cultural backgrounds, personal circumstances and behaviors in addition to finding themselves in well-defined academic tracks which all in some way shape their personal academic goals and have an impact on their academic performance. I have had the opportunity since Spring 2018, over a span of fourteen consecutive semesters, to teach undergraduates mathematics in courses at QC ranging from college algebra (Math 115) to probability and statistics (Math 241) and everything in between. Typically, on the first day, I have them

introduce themselves and at the very least answer two questions: 1) what has been your experience with learning mathematics? and 2) what is your goal in taking the given course? I get a range of responses in return that give me some sense of where these students come from and where they potentially are heading toward. However, I recognize that this first encounter could be more useful if the information that they come into the classroom with may be utilized to preemptively gauge their potential performance for the semester.

The exploration of the interplay that exists between the educational processes and student performance has been an ongoing point of interest in the field of education. Due to the recent advancement of computational technologies coupled with the proliferation of data creation and gathering processes, this field, known as *educational data mining* (EDM), of analyzing student data along with statistical methods to solve problems in education has gained more traction and has rapidly grown over the past twenty years. As both a student and educator of mathematics, certain questions arise: How feasible and effective is it to predict a students' performance in a mathematics course before they set foot in class? What are the major factors that contribute to a student's success or failure in a university math class? Is there an opportunity to implement corrective measures throughout the semester to address any crucial deficiencies in time to ensure a passing grade? In this paper, I aim to investigate both theoretical and concrete answers for these questions through investigating the efficacy of predicting a student's final letter grade in a university mathematics class under a predictive linear modeling framework.

II. Modeling Framework

The Theoretical Setup

A *model* is an attempt by us humans to generalize a truth, predict an event, explain a system that can be observed and/or experienced in the world we live in. We refer to these real-world truths, events and/or systems as *phenomena*. In short, *models* are “approximations

to reality.” George Box, a well-known British statistician, once quipped that “all models are wrong, but some are useful.” The utility of models are two-fold: 1) models are useful for prediction, and 2) models are useful for providing interpretable insights (explanation) to a phenomenon of interest. But how do we confidently verify the precision and accuracy of any model that we come up with? First, there is the need for *empirical* data which can be collected, measured, and stored. We are able to make most sense of the data aggregated from the real-world by either making use of its inherent numerical nature or converting it numerically when it is not. The numerical data is then assessed with *metrics* spawning the need for mathematical understanding. Our model then becomes a *mathematical* one that takes numerical inputs of data and produces a unique numerical output which, as a result, can be mathematically analyzed and evaluated for its reliability or lack thereof in its approximation to the real-world phenomenon. This modeling paradigm provides the framework to look into my phenomenon of interest: a QC student’s performance measured by a final letter grade in a given mathematics class.

To construct models that produce predictions of any sort requires to soberly make plain and come to terms with the reasons why any model is “wrong” in the first place, especially mathematical ones. For starters, our model will be represented by a mathematical function denoted in general by f . A function is a rule that assigns an element of a set of inputs also known as independent variables, denoted by $\mathcal{X}$, to a single element, y , of a set of outputs, $\mathcal{Y}$, summarized succinctly by $f : \mathcal{X} \rightarrow \mathcal{Y}$ where $y = f(x)$ and $y \in \mathcal{Y}$ and $x \in \mathcal{X}$ ($\in$ denotes “belongs to”). For simplicity’s sake y will be a real-valued number, i.e., $\mathcal{Y} = \mathbb{R}$ where $\mathbb{R}$ denotes the set of all real numbers. However, the set of our inputs for our mathematical model is not as straightforward to ascertain and in defining our set of inputs is exactly where we incur our first fundamental source of error in any proposed model we make.

Causal Drivers & Output Space

We first posit that a “true” function exists which perfectly explains our phenomenon. Call this function t . This function will theoretically be dependent upon the *causal drivers* of our phenomenon of interest, also called the *proximal causes* – the true causes/factors that precisely explain our phenomenon denoted by $z_1, \dots, z_t$. We can now express our phenomenon in function notation as

$$y = t(z_1, \dots, z_t).$$

The fundamental problem with our theoretical framework thus far is that t can never be known in the real-world. Otherwise, George Box’s aforementioned quote would be a facetious lie and modeling would not be as necessarily useful. Secondly, we cannot exhaustively and assuredly known every possible causal driver, $z_1, \dots, z_t$, of our phenomena. And even if we even did know each of the z_i ’s we still would not be able to collect nor measure them all. Our modeling enterprise thus becomes both an art and a science of approximating t the true function representing our phenomenon of interest.

The t in my context refers to the phenomena of a student’s performance in a QC undergraduate 100 - 200 level mathematics class. For specificity, I will be looking into the student’s performance in my favorite class to teach – Linear Algebra (Math 231). The output space $\mathcal{Y}$ is the set of all outputs of my phenomenon, meaning all possible final letter grades, namely $\mathcal{Y} = \{A+, A, A-, B+, B, B-, C+, C, C-, D+, D, F\}$. Now, I will convert these letter grades to their respective numerical equivalents out of a 4.0 scale, namely $\mathcal{Y} = \{4.3, 4.0, 3.7, 3.3, 3.0, 2.7, 2.3, 2.0, 1.7, 1.3, 1.0, 0.0\}$. The hard part is to identify every causal driver that explains how a student ultimate assignment of a final letter grade. One can make a possible conjecture for the causal drivers:

- z_1 = a student’s intellectual capacity at the time of taking the course
- z_2 = a student’s inclination for learning mathematical ideas

- z_3 = the student's logical reasoning skills
- z_4 = how responsible a student is to meeting deadlines for assignments
- z_5 = the quality of a student's study habits

These proposed proximal causes may sound plausible yet obviously not exhaustive. The proximal causes are specific to the student and are quite qualitative in nature which make it quite hard and nearly impossible to measure and/or collect. One can propose other proximal causes that may be related to the student's environment such as the number of daily hours of sleep a student has or how well a student takes notes in class. One can even argue of more ancillary factors such as a student's determination to figure out a mathematical problem when facing difficulty or the quality of mathematical instruction a student has received over the course of their academic careers. The point is that though all of the above are potential indicators for a student's performance in a QC mathematics class they are essentially impossible to measure and to keep track of for each potential student. We need more concretely defined variables that can be measured and can serve as a "proxy" to the phenomenon's causal drivers.

Proxies, Errors & Learning

Since we cannot enumerate all of the causal drivers for our phenomenon nor can we measure them all, the next best thing is to define variables that proxy the value of the z 's as best as possible which are called *proxies* also known as *features* denoted by $x_{.1}, x_{.2}, \dots, x_{.p}$. The number of features is denoted by p . Now our t is best approximated by f which will be the approximated function of our model based upon the proxies. We then have our first source of error denoted by δ known as *error due to ignorance* because of our inherent ignorance to the actual true proximal causes. Our phenomenon undergoes a slight change in notation:

$$y \neq f(x_1, x_2, \dots, x_p) \implies y = f(x_1, x_2, \dots, x_p) + \delta \quad (\delta = t - f)$$

Our goal for making a model of our phenomenon has been reformulated to finding f , the *target function* which approximates t . To decrease the error due to ignorance one has to increase the number of relevant features, p . The downside here is that we do not know all relevant features that pertain to the phenomenon and as a result we can never fully know f either. This is we find another source of error and another need for approximation in our pursuit for finding the “best” model. The science to modeling is to know what features are deemed as relevant that explain the phenomenon and to figure out how many of these features are there. The art of modeling is to figure out what type of functions are suitable to approximate f . The approximation of f based upon the p number of relevant features is what we will loosely call *learning*.

To approximate our target function f we “learn” from historical data also known as *training data* that has been previously collected. This empirical approach for approximating f is called *supervised learning*. For supervised learning we need three ingredients:

- (i) The training Data denoted by $\mathbb{D}$ consists of

$$\mathbb{D} = \{\langle \mathbf{x}_1, y_1 \rangle, \langle \mathbf{x}_2, y_2 \rangle, \dots, \langle \mathbf{x}_n, y_n \rangle\}$$

where n represents the number of historical examples, each proxy $\mathbf{x}_i$ has p measurements $\mathbf{x}_i = \begin{bmatrix} x_{i,1}, x_{i,2}, \dots, x_{i,p} \end{bmatrix}$, each i is known as a *unit* or *record*, and each y_i is the known *label* or *response* of the i^{th} unit. We employ an extremely useful way to store numerical information into rows and columns by matrix $\mathbf{X}$ where each row is the proxy $\mathbf{x}_i$. As a result, $\mathbf{X}$ is an $n \times p$ matrix and the training data $\mathbb{D}$ can be succinctly summarized as $\mathbb{D} = \langle \mathbf{X}, \mathbf{y} \rangle$ where $\mathbf{y}$ is a $n \times 1$ column vector of responses y_i .

- (ii) A set of candidate functions to approximate f denoted by $\mathcal{H}$
- (iii) An algorithm, a specific sequence of instructions, $\mathcal{A}$, that takes $\mathbb{D}$ and $\mathcal{H}$ as inputs and outputs the optimal function in the set $\mathcal{H}$ which we call g . i.e, $g = \mathcal{A}(\mathbb{D}, \mathcal{H})$

The concoction of the above ingredients in supervised learning for approximating f immediately leads the rise of two additional sources of error. In the candidate set of functions $\mathcal{H}$ that will be used to approximate f has a theoretical best function called h^* which is the function in $\mathcal{H}$ with the lowest feasible error. Inherently, $h^* \neq f$ and so the error incurred by $f - h^*$ is known as *misspecification error*. This kind of error inherently arises from not knowing the type of function that f is and upon specifying the type of functions that are in $\mathcal{H}$. And we are able to reduce misspecification error by redefining $\mathcal{H}$ to be more expansive in the types of functions it contains. The last type of error we have comes from within the candidate set $\mathcal{H}$. We ultimately will never know h^* either since we will not have the complete number of features nor will we have access to all of the number of historical examples that would help us find the function in $\mathcal{H}$ with the lowest error. That is why in the supervised learning step (iii) we arrive at the optimal function g in $\mathcal{H}$ which is considered the best function after h^* since it is the function with the lowest error given the available set of features and historical data. The error attributed by $h^* \neq g$ or $h^* - g$ is known as *estimation error*. By letting $\mathbf{x}_i = \begin{bmatrix} x_{i,1}, x_{i,2}, \dots, x_{i,p} \end{bmatrix}$ we can summarize all three errors by the following:

$$y_i = g(\mathbf{x}_i) + \underbrace{(h^*(\mathbf{x}_i) - g(\mathbf{x}_i))}_{\text{estimation error}} + \overbrace{(f(\mathbf{x}_i) - h^*(\mathbf{x}_i))}^{\text{misspecification error}} + \underbrace{\delta}_{\text{ignorance}}$$

The sum of all three errors is called a *residual* denoted by e_i . Clearly defining the components of supervised learning and sources of error for approximating f sets the stage for giving a concrete frame to build a model upon for a QC's student performance by first establishing the features that will give rise to the training data $\mathbb{D}$, the types of functions we will be sifting through in $\mathcal{H}$ and the procedure, $\mathcal{A}$, by which we select our best modeling function g .

Feature Selection

Now that we have laid down the theoretical framework needed for the implementation of supervised learning in the modeling process we can start selecting the features that will be used in amassing our training data set. The selection of the features should be gauged by its relevancy to student's performance in a mathematics class and by the ability for features to be measured/collected preferably in a numerical fashion. Consider the following features:

- (a) x_1 = GPA of past mathematics classes
- (b) x_2 = # of hours dedicated to homework a week
- (c) x_3 = year at QC when taking class (freshmen, sophomore, junior, or senior)

The first two variables are numeric values. The GPA's of past mathematics classes could theoretically be searched and collected in the database that QC stores its students' grades in. I can also go into my CUNYFirst account and I find all of my past students final letter grades in my previous Math 231 classes for example. I mostly have used the homework learning management system WebAssign for the homework and I can find the time it took for each student to complete each homework assignment and could average all of the times together. The x_3 is an ordinal categorical type of feature which I can "dummify" them by saying that a freshman is labeled a numerical value of 1 and a sophomore is labeled a numerical value of 2 and so on. For now these are the relevant features that can be collected as training data. I have taught Math 231 in the past 4 semesters of classes sizes that roughly average around 30 students, so I can create a training set matrix $\mathbf{X}$ of 120 rows and 3 columns. The three columns refer to the features we have specified above and each of the 120 rows represents the record of an individual student. The corresponding column vector of labels, $\mathbf{y}$ will be the final letter grade in its equivalent numerical value that they received for the class.

III. Model Specification & Prediction

After specifying the features of our training data and detailing the practical ways that we can collect this data $\mathbb{D} = \langle \mathbf{X}, \mathbf{y} \rangle$, we need to make a decision on what our candidate set $\mathcal{H}$ will end up being. There are multiple possibilities for the types of functions to choose from. Since our response comes from the set of real numbers, our candidate set will be made up of all linear functions namely

$$\mathcal{H} = \{ \mathbf{w} \cdot \mathbf{x} : \mathbf{w} \in \mathbb{R}^4 \} = \{ w_0 + w_1x_1 + w_2x_2 + w_3x_3 : w_0, w_1, w_2, w_3 \in \mathbb{R} \}.$$

There is a subtle point that has to be clarified surrounding the intercept value, w_0 , of our linear set of candidate functions. This intercept value should give us value that would be make sense if we do not know any information from our features. Our training data matrix $\mathbf{X}$ also known as our *design matrix* will then have an augmented column of 1's along its left-most column. So our new design matrix $\mathbf{X}$ will still have 120 rows but 4 columns instead of 3. The algorithm, $\mathcal{A}$, that we will implement will minimize our error given by an objective function. Recall that the residuals e_i is the error made by the function g in $\mathcal{H}$ and the known label y_i . We call $\hat{y}_i = g(\mathbf{x}_i)$ our predicted value. The sum of all the squared residuals will be known as our objective function that we are attempting to minimize which is called the *sum of squared errors* (SSE). It is also known as the L2 loss and is often the default objective function that we are trying to minimize (or maximize):

$$\text{SSE} = \sum_{i=1}^n e_i^2 \implies \mathcal{A} : \equiv \text{argmin} \{ \text{SSE} \}$$